

Technology evolution

In 1921, Ernst Sachs, founder of ERSAs (a name consisting of the beginning letters of his first and last names), developed the first electric and mass-produced soldering iron for the industry. From then on, there have been many advancements in this field. Soldering iron was soon replaced with temperature-controlled soldering stations. Nowadays, SMD rework stations are the latest trends. However, because of their good quality and lower costs, soldering stations are still widely used.

Talking of technology trends, there is a shift towards digital, ESD-safe and lead-free soldering stations.

Analogue vs digital

A soldering station can be either analogue or digital. Further, "in analogue there are temperature-controlled and wattage-controlled stations, whereas in digital you have auto temperature-controlled option. The main difference is that in analogue stations, fluctuation in the temperature is higher, whereas in digital you can have a constant temperature. Also, in digital, you get many other options like auto sleep. Today, analogue stations are most widely used due to their lower price," informs Mukesh Rajput, CTO, Max Technology & Co.

Analogue units have knobs to control (increase or decrease) the temperature. There is no display to show the current set temperature. Also, the temperature setting in these units is not very accurate.

On the other hand, digital units have settings to control the temperature digitally. These also have a display that shows the current set temperature. They offer better accuracy but cost more than their analogue counterparts.

Says S.K. Jain, managing director, Sumitron Exports, "Both analogue and digital types are used in production lines as well as R&D. Repair centres also use both types depending on the application."

Several systems come with ana-



Hakko FX-951 soldering station

logue control but digital display. Often, customers misunderstand them as digital stations. Digital stations are the stations where control is digital apart from the display. These bring a lot of benefits to control various process parameters.

"The latest soldering stations have their tip integrated with the heating element, allowing constant temperature and quick heat generation and recovery. As air (a bad conductor of heat) is absent, end users are able to convert most of the energy in work, which results in increased productivity. Even lead-free soldering, which needs a high temperature, can be done at a lower temperature. This leads to better solder joint, longer tip life and safe electronic components," informs Ajay Gupta, director-technical, Advance Tech Services.

"There has been a definite change towards digital stations as these help in efficient working temperature, which is needed when working with temperature-sensitive components," says Kwatra.

There are a wide variety of equipment available in the market depending on the use, i.e., whether the

user wants to work on plate-through hole (PTH) or SMD components. The quality and features of equipment available may vary. The prices have come down drastically in last five years.

OK International's Metcal MX-5000 soldering and rework stations have a built-in power in-

dication meter with digital display and bar graph that dynamically provides the operator with feedback on the status of the soldering operation.

ESD safe and non-ESD safe

Electrostatic discharge (ESD) is the sudden and momentary electric current that flows between two objects at different electrical potentials caused by direct contact or induced by an electrostatic field. Failure to protect sensitive electronic components from ESD can damage the component to the extent of failure.

Products which meet the following specifications are called ESD-safe:

1. The unit is grounded and the resistance between the tip and ground is less than 2 ohms
2. The tip-to-ground potential does not exceed 2 mV
3. ESD-protective materials are used

Soldering stations that guarantee that the static energy from a person's body (hands, etc) will be 'dumped' to ground via the plastic soldering iron handle or solder station plastic housing are called ESD-safe soldering stations. The solder station essentially discharg-



Production and repairing centres in electronics and telecom industries are the areas where demand for soldering and desoldering stations is the highest—Sumeet Chadda, director, Blue Sky Systems